

Problem Title: The Lost Codex Cipher

Background

You are a researcher. While organizing a collection of recently unearthed ancient manuscripts, you discover a booklet that records a mysterious numerical pattern.

The booklet describes a special sequence where the i -th number is given by the formula:

$$P(i) = a \cdot i^2 + b \cdot i + c$$

Here, a , b , and c are three positive integers written on the first page of the booklet.

The crucial instruction at the end of the booklet states that **the product of the first n numbers in this sequence** is the key to unlocking the final secret of the codex. Since n can be very large, the resulting product may be an extremely large integer.

Your task is: given n , a , b , and c , compute the product of these n numbers and output the exact value of this huge integer.

Input Format

A single line containing four space-separated integers:

n, a, b, c

Output Format

Output a single integer representing the exact value of:

$$P(1) \times P(2) \times \cdots \times P(n)$$

Sample

Sample Input 1

```
5 1 2 3
```

Sample Output 1

```
1218888
```

Constraints

- $1 \leq n \leq 20000$
- $1 \leq a, b, c \leq 10000$

Notes

You are **not allowed** to use any built-in types or third-party libraries that directly support arbitrary-precision multiplication (e.g., Java's `BigInteger`).